Author-designed CNN variants: input to output

Input: 9 × 9 × 4 HLS patch · four bands · 30 m
Output: survey z (m), relative to the site's port zero

CNN	3 × 3 conv encoder · 32 → 64 → 128 Global average pooling → dense head → scalar z
RatioCNN	CNN encoder + centre ln(B02/B03) Dense head → scalar z
CNN–BiLSTM	CNN encoder → four spatial tokens Bidirectional LSTM → scalar z
DenseCNN	CNN encoder → transpose-convolution decoder 9 × 9 z field → score centre cell
DepthBinCNN	CNN encoder → z regression + 3-bin head Bins use the same survey z labels

Baselines: Stumpf blue–green log-ratio and random forest
Both are fitted within each training fold